

RETAIL BUSINESS INTELLIGENCE

& Customer Analytics

An end-to-end SQL analytics project on the Olist Brazilian e-commerce dataset — from raw data to a decision-ready Power BI dashboard.

PostgreSQL 18

Power BI

35+ Business Questions

9 Normalized Tables

6 Analytics Views

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Executive summary

Turning 100K+ raw e-commerce records into an analytics-ready decision layer

Objective

Build a full analytics pipeline on Olist's Brazilian e-commerce dataset — from schema design through data-quality validation, SQL-based business analysis, and a Power BI reporting layer — to answer real revenue, customer, and operations questions.

Scope

9 relational tables, 35+ business questions across revenue, sellers, customers and logistics, 6 production-style analytics views, and a 7-page interactive dashboard.

R\$16.01M

Total sales analyzed

110K

Total orders

100K

Unique customers

77%

Positive review rate

Data pipeline & tech stack

Raw CSVs to a decision-ready dashboard, in five stages



Backend — PostgreSQL 18

- 9 normalized tables with primary/foreign keys
- CTEs, window functions (RANK, ROW_NUMBER, NTILE, LAG)
- 6 reusable analytics views for reporting
- Systematic null, duplicate and range checks

Frontend — Power BI

- 7-page interactive dashboard
- Sales, sellers, customers, payments, reviews
- Year slicers and drill-through order details
- Custom visuals: tornado, radar, timeline charts

Data model — 9 normalized tables

Star-like schema linking orders, customers, sellers, products and payments

■ customers

Customer id, unique id, city, state, zip prefix

■ orders

Order status and full purchase-to-delivery timestamps

■ order_items

Line items: product, seller, price, freight value

■ products

Category, dimensions, weight, photo count

■ sellers

Seller id, city, state, zip prefix

■ payments

Payment type, installments, value per order

■ reviews

Review score, comments, response timestamps

■ geolocation

Lat/Lng by zip code prefix

■ category_translation

PT to EN product category mapping

Data quality & governance

Every table validated before a single business question was asked

- ✓ Row counts confirmed across all 9 tables before analysis began
- ✓ Missing-value scan on category, weight, dimension and delivery-date fields
- ✓ Duplicate primary key checks on customers, orders, products and reviews
- ✓ Dataset time span confirmed (first order to last order)
- ✓ Order status distribution profiled to isolate delivered vs. cancelled orders
- ✓ Customer distribution by state validated for geographic coverage

Why this matters

Skipping data-quality checks is the fastest way to produce confident-looking, wrong numbers. This phase caught null delivery dates, category gaps and duplicate keys before they could distort revenue and delivery-time calculations downstream.

Business KPI overview

The headline numbers behind the full analysis

R\$16.01M

Total sales

R\$120.76

Average order value

3,095

Active sellers

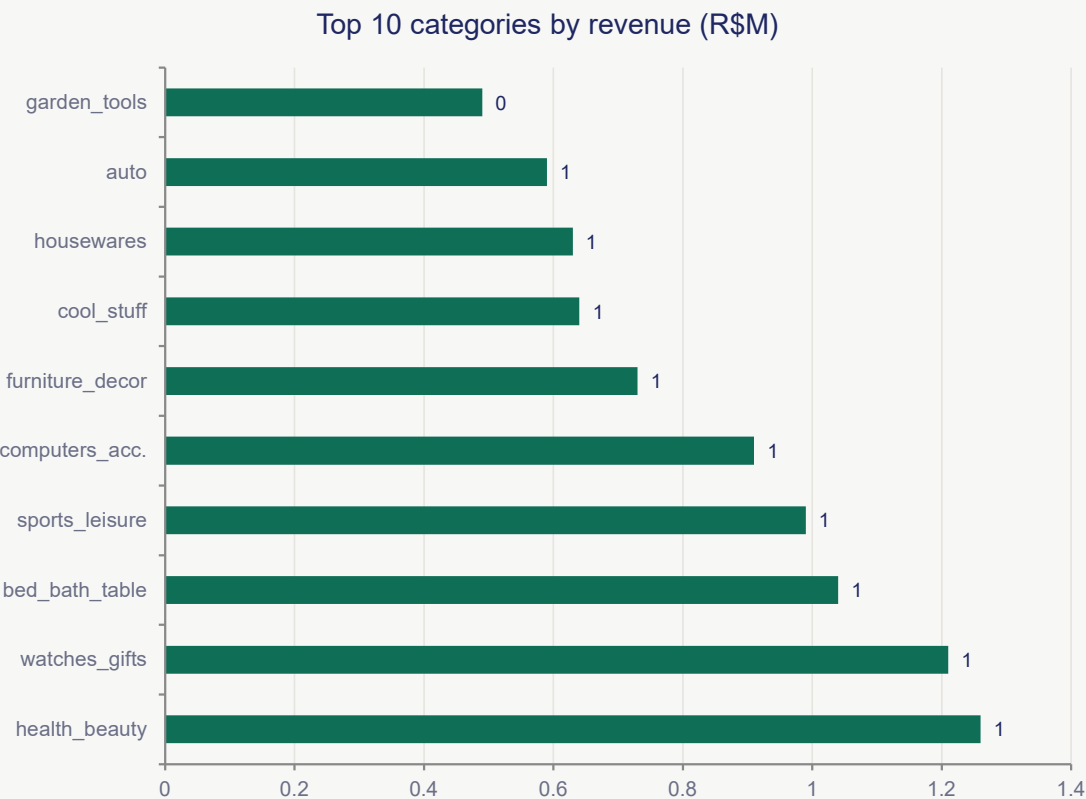
R\$4,332

Average sales per seller

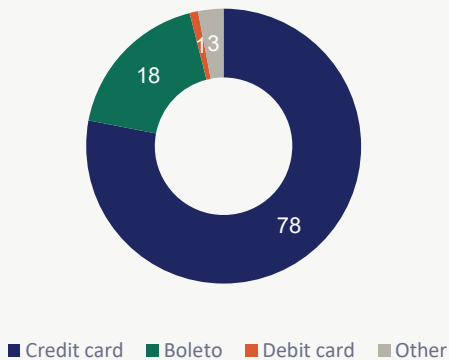
Sourced from the executive-summary and seller-overview analytics views (vw_executive_summary, vw_state_performance) built in the SQL layer — not hand-computed, so every number in the dashboard traces back to a query.

Revenue insights

Category mix and payment behavior across 110K orders



Payment method share (%)



Top state — Sao Paulo (SP)

R\$5.13M in sales, ~40% of all revenue — the single largest concentration in the dataset.

Advanced SQL — window functions in action

Four techniques used to move beyond simple aggregation

ROW_NUMBER() — Top products per category

```
ROW_NUMBER() OVER (  
  PARTITION BY category  
  ORDER BY products_sold DESC  
) AS ranking
```

Ranks every product within its own category so the top 3 sellers per category can be isolated in one pass.

RANK() / DENSE_RANK() — Seller leaderboard

```
RANK() OVER (  
  ORDER BY revenue DESC  
) AS seller_rank
```

Ranks all 3,095 sellers by revenue, comparing RANK (with gaps) against DENSE_RANK (no gaps) for tied values.

SUM() OVER(ORDER BY) — Running revenue

```
SUM(revenue) OVER (  
  ORDER BY month  
) AS cumulative_revenue
```

Builds a running monthly total without a self-join, powering the cumulative-growth view of the business.

SUM() OVER() — Revenue contribution %

```
100 * revenue /  
  SUM(revenue) OVER ()  
  AS contribution_pct
```

Expresses each category's revenue as a share of total revenue in a single query, no post-processing needed.

Customer analytics — RFM & segmentation

Turning transaction history into actionable customer groups

RFM metrics computed per customer

Recency `MAX(order_purchase_timestamp)`

Frequency `COUNT(order_id)`

Monetary `SUM(payment_value)`

Customer value segments

Low value

< R\$100

Medium value

R\$100 – 500

High value

R\$500 – 1,000

Premium

> R\$1,000

Additional customer-level analysis

- Repeat vs. one-time customers, using `customer_unique_id` to correct for re-registered accounts
- Top 1% customers isolated with `NTILE(100)` for a targeted retention list
- Customer lifetime value (CLV) per unique customer
- Average orders per customer and full order-count distribution
- States ranked by share of repeat customers

Delivery performance & review correlation

Connecting logistics speed to customer satisfaction

1 Fastest / slowest delivery states

Average delivery days computed per customer state to flag logistics bottlenecks by geography.

2 Delivery time vs. review score

Average delivery duration grouped by review score (1–5) to test whether slower delivery drives lower ratings.

3 Revenue by weekday

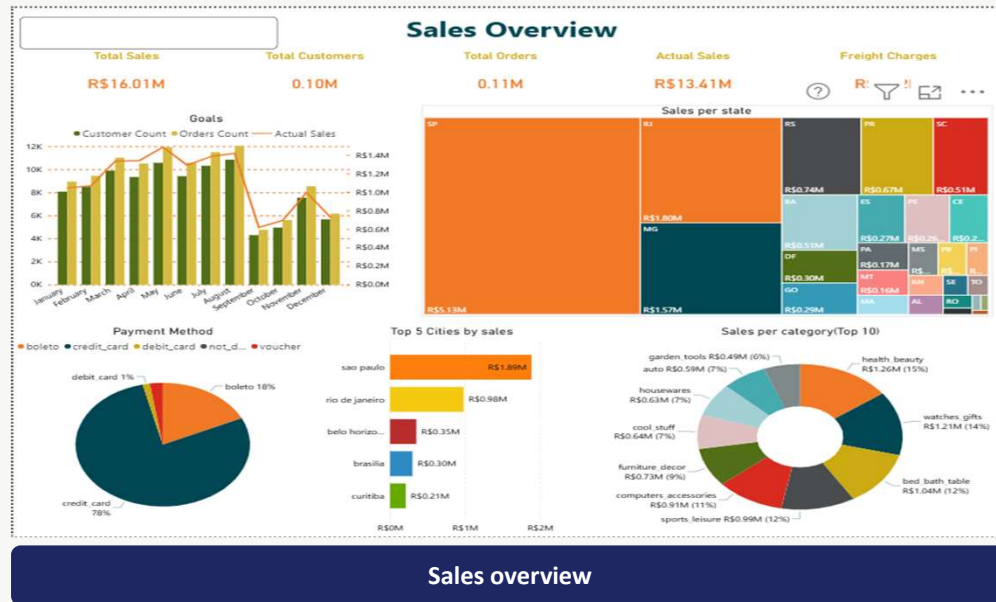
Sales and order volume broken down by day of week to spot purchasing rhythm.

4 Revenue by hour

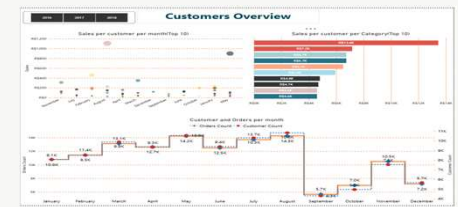
Purchase-hour distribution used to identify peak ordering windows across the day.

Power BI dashboard walkthrough

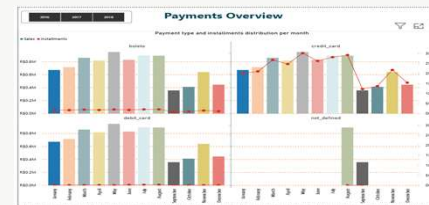
SQL views brought to life across five focused pages



Sellers overview



Customers overview



Payments overview



Reviews overview

Full interactive file: [Brazil_Ecommerce_Analysis.pbix](#) | Year slicers (2016–2018) and drill-through order detail on every page.

Key insights & recommendations

What the data says, and what to do about it

Key findings

SP drives ~40% of total sales, creating regional concentration risk.

Sales drop sharply Sep–Dec — likely a dataset coverage gap, not a real demand decline.

Credit card is 78% of payments; boleto usage stays steady year-round.

Top 5 categories drive most revenue; a long tail of 60+ categories underperforms.

Recommendations

1 Expand seller acquisition outside SP to reduce single-state dependency.

2 Investigate the Sep–Dec sales cliff before treating it as a seasonal trend.

3 Prioritize the top 5 categories for inventory and marketing spend.

4 Pair review-score analysis with delivery time to target complaint drivers.

Data note: Late-2016 records are sparse in the source dataset — treat that period's dips as data-coverage gaps, not real demand trends.

Skills demonstrated

SQL

- CTEs & subqueries
- Window functions (RANK, ROW_NUMBER, NTILE, LAG)
- Views for reusable reporting
- Data-quality auditing

Analytics

- RFM & customer segmentation
- Cohort-style repeat-purchase analysis
- Revenue attribution & contribution %
- Delivery-to-review correlation

BI & delivery

- Power BI dashboard design
- DAX measures & slicers
- Insight-driven storytelling
- Version-controlled SQL scripts

Thank you — happy to walk through any query or dashboard page in detail.